COMPUTER PROGRAMMING

LAB 4

LAB TITLE

Switch Statement, Break and Continue in C++1. Objective

The main objective of this lab is to understand how the **switch statement** works in C++. In this lab, students learn how a program selects one option from multiple choices using switch. We also understand the role of the **break statement** in stopping program execution and the **continue statement** used inside loops. This lab also helps students create simple **menu-driven programs**.

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2. Theory

In C++ programs, sometimes we need to make decisions based on user input. When a program needs to compare one variable with several possible values, the **switch statement** is very useful. It is often used instead of multiple **if-else statements**.

Inside the switch statement, different choices are written as **cases**. The program checks the value entered by the user and runs the matching case.

The **break statement** stops the execution after a case runs. If break is not used, the program will continue executing the next cases.

The **default statement** runs when the user input does not match any case.

The **continue statement** is used in loops to skip the current iteration and move to the next iteration.

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3. Software

- DEV-C++ IDE
- C++ Compiler (included in DEV-C++)

Program 1: Day Name Using Switch

Algorithm

1. Start the program
2. Declare an integer variable for day number
3. Ask the user to enter a number from **1 to 7**
4. Use switch statement to check the number
5. Display the corresponding day name
6. End the program

Code

```
#include <iostream>
using namespace std;

int main()
{
    int day;

    cout << "Enter day number (1-7): ";
    cin >> day;

    switch(day)
    {
        case 1: cout << "Monday"; break;
        case 2: cout << "Tuesday"; break;
        case 3: cout << "Wednesday"; break;
        case 4: cout << "Thursday"; break;
        case 5: cout << "Friday"; break;
        case 6: cout << "Saturday"; break;
        case 7: cout << "Sunday"; break;
        default: cout << "Invalid day number";
    }

    return 0;
}
```

Program 2: Grade Evaluation Using Switch

Algorithm

1. Start the program
2. Declare a character variable for grade
3. Ask the user to enter grade
4. Use switch statement to check grade
5. Display message according to grade

6. End the program

Code

```
#include <iostream>
#include <cctype>
using namespace std;

int main()
{
    char grade;

    cout << "Enter your grade (A, B, C, D, F): ";
    cin >> grade;

    grade = toupper(grade);

    switch(grade)
    {
        case 'A': cout << "Excellent"; break;
        case 'B': cout << "Very Good"; break;
        case 'C': cout << "Good"; break;
        case 'D': cout << "Pass"; break;
        case 'F': cout << "Fail"; break;
        default: cout << "Invalid Grade";
    }

    return 0;
}
```

Program 3: Unit Conversion Menu Using Switch

Algorithm

1. Start the program
2. Display menu options
3. Take user choice
4. Ask for value
5. Perform selected conversion
6. Display result
7. End the program

Code

```
#include <iostream>
using namespace std;
```

```
int main()
{
    int choice;
    double value;

    cout << "1 -> Kilometers to Meters\n";
    cout << "2 -> Meters to Centimeters\n";
    cout << "3 -> Kilograms to Grams\n";
    cout << "4 -> Celsius to Fahrenheit\n";

    cout << "Enter your choice: ";
    cin >> choice;

    cout << "Enter value: ";
    cin >> value;

    switch(choice)
    {
        case 1:
            cout << "Meters = " << value * 1000;
            break;

        case 2:
            cout << "Centimeters = " << value * 100;
            break;

        case 3:
            cout << "Grams = " << value * 1000;
            break;

        case 4:
            cout << "Fahrenheit = " << (value * 9/5) + 32;
            break;

        default:
            cout << "Invalid Choice";
    }

    return 0;
}
```

Conclusion

In this lab, we learned how to use the **switch statement** for making decisions in C++. We also understood the importance of the **break statement** in controlling the flow of the program. This lab helped us create simple **menu-driven programs** and improved our understanding of conditional structures in C++.